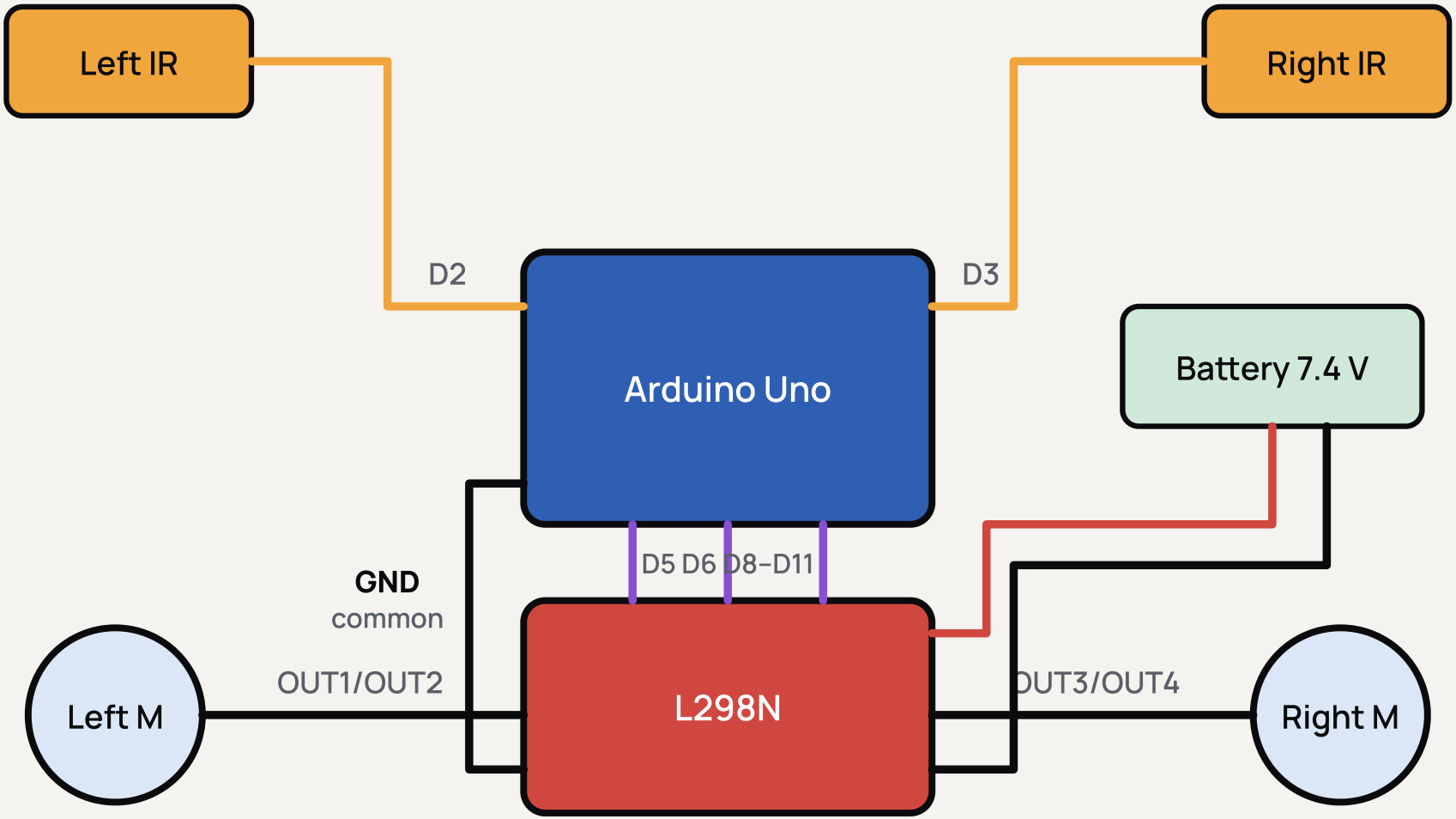
Wiring diagram

Arduino Uno + L298N + two IR sensors

Wiring: battery → driver → motors, Arduino → driver



Signal	Arduino pin	Name in code	Type
Left IR sensor OUT	**D2**	`LEFT_IR`	Digital input
Right IR sensor OUT	**D3**	`RIGHT_IR`	Digital input
L298N ENA (left motor speed)	**D5**	`ENA`	PWM output (~)
L298N ENB (right motor speed)	**D6**	`ENB`	PWM output (~)
L298N IN1 (left motor direction)	**D8**	`IN1`	Digital output
L298N IN2 (left motor direction)	**D9**	`IN2`	Digital output
L298N IN3 (right motor direction)	**D10**	`IN3`	Digital output
L298N IN4 (right motor direction)	**D11**	`IN4`	Digital output

Power: each IR sensor VCC → 5V and GND → GND. Battery + → switch → L298N 12V; battery − → L298N GND; L298N GND → Arduino GND (common ground). Left motor → OUT1/OUT2 (motor A); right motor → OUT3/OUT4 (motor B).

Before power: batteries out while wiring · all grounds joined (battery −, L298N GND, Arduino GND, sensor GND) · ENA/ENB jumpers removed · an adult has checked every wire.